Controls, checkpoints and troubleshooting

Plan reference materials, record recovery by stage and investigate unexpected results.

Working document. Red **TO COMPLETE** fields identify missing or conflicting author inputs. Complete the fields affecting a procedure before using it at the bench. Recorded settings are not a claim of experimental optimization.

In this document

01 Plan the checks before the assay
02 Control plan
03 Qualify a small run
04 Progression checkpoints
05 Cells, pairing and sorting
06 Target recovery and sequencing
07 Assignments and interaction evidence

01 Plan the checks before the assay

This guide combines source-identified problems with diagnostic recordkeeping proposed in the author-review draft. The notes do not supply generally validated numerical acceptance limits. Record the reference material, expected result and acceptance rule for each checkpoint before the full run. [D1 sections 2.4, 9]

TO COMPLETE - C13
Provide named, experimentally supported positive and specificity/background controls, replicate design, reference-run QC distributions and checkpoint acceptance rules. Attach evidence supporting each expected result.

Keep the measurement and its interpretation separate
A co-positive sorter event, shared partition or large read count does not by itself establish a specific TCR-pMHC interaction. Input-library aliquots measure representation; they do not directly measure background pairing. [S1 pp.1-4]

02 Control plan

The following controls are recommended additions for reproducibility, not a claim that they were demonstrated in the source notes. Assign each control a sample ID, material identity, planned checkpoint and expected result.

Control or check	What to assess	Retain
Unstained and appropriate single-color controls	Fluorescence background and compensation/unmixing.	Control identities, raw FCS files and instrument settings.
Each input population separately	Marker specificity, background co-positivity and sample identity.	Gate hierarchy, denominators and observed distributions.
Author-validated positive TCR-HLA-peptide combination	Recovery of an expected association through the complete procedure.	Material versions, supporting evidence and observed result.
Matched specificity/background control	Nonspecific association and event recovery in the relevant context.	Control rationale, expected behavior and measured background.
Post-sort composition and recovery	Intended pairs, larger aggregates, losses and dissociation.	Assessment method, composition and separate event/cell/pair counts.
PCR no-template and appropriate process blanks	Contamination during handling or amplification.	Sample/branch allocation, amplification results and raw traces.
Independently prepared assay replicates	Repeatability across preparations.	Preparation IDs, dates, conditions and separate analysis outputs.

A model score does not substitute for an experimental control. Repeated sequencing of one library is not a biological replicate. Keep each additional capture reaction and indexed PCR branch identifiable in the sample map.

03 Qualify a small run

1. Run the source-supported flow pilot with the intended libraries and record the sampling time, controls, gate hierarchy and gate denominator. Use the measured yield to plan collection. [S1 p.1]
2. As a recommended next qualification, carry a small run through crosslinking, sorting, capture, targeted PCR and analysis. A flow-only pilot does not establish recovery through those later stages.
3. Compare observations with the defined control results and acceptance criteria. Locate losses at the stage where they occur, retaining the raw measurements and failed samples.
4. Before scaling, record changes in cell concentration, total volume, vessel and shaker geometry, timing and recovery. More cells or co-positive events alone do not establish better specificity.

TO COMPLETE - C04
Supply the qualified pairing conditions and scale range, including medium, concentration, temperature/CO_2, vessel geometry and timing. Keep pilot observations separate from release acceptance limits.

TO COMPLETE - C06
Attach portable sorting settings and a post-sort composition/recovery method. Define how intact intended pairs are distinguished from other collected events.

04 Progression checkpoints

Checkpoint	Evidence to review	Completion item
Constructs	Correct complete sequences, primer compatibility, known library contents and representation.	TO COMPLETE: C01/C08 - qualification records.
Cell libraries	Cell identity, culture quality, expression, viability/recovery and post-selection or post-thaw representation.	TO COMPLETE: C01/C02 - criteria and results.
Flow pilot	Controls/settings, explicit gate denominator, observed yield and feasible collection plan.	TO COMPLETE: C04/C06/C13 - criteria and files.
Post-sort	Composition, recovered material, transfer losses and unit-labeled capture loading calculation.	TO COMPLETE: C06/C07 - method and calculation.
Target recovery	Four required PCR2 branches, blank results and construct-specific fragment identities.	TO COMPLETE: C09/C10 - programs and criteria.
Sequencing	Branch assignments and sampling against predeclared criteria; raw files and logs retained.	TO COMPLETE: C11/C12 - run and analysis criteria.
Interpretation	Controls, replicate results, reproducible assignments and stated interaction-calling method.	TO COMPLETE: C12/C13 - method and reference results.

For each checkpoint, record the planned criterion, actual result, file location, decision and any deviation. When a result falls outside the released criterion, document the affected stage and resolution before treating its downstream output as qualified.

Use explicit counting units
Keep cells, acquired sorter events, co-positive sorter events, collected events, intact pairs, loaded material and recovered partitions separate. At analysis, distinguish reads from molecules and supporting partitions. Report molecule support only if UMI retention and processing have been established. [S1 pp.1-2]

05 Cells, pairing and sorting

Start with the checks below when results differ from the reference run. These checks do not authorize unvalidated changes to concentrations, incubation conditions or gates.

Observation	First checks	Record or next action
Poor cell recovery or depleted library	Cell health, measured titer basis, actual dose, selection timing, culture bottlenecks and freeze/thaw history.	Reassess upstream qualification. Expansion alone does not restore missing library members. [S4; S5]
Violet signal saturates or overlaps negatives	Product and stock identity, reconstructed dilution, instrument settings and controls.	Resolve the missing stock specification before changing the staining recipe. [S7 p.1; C03]
Few co-positive events	Input expression/representation, concentrations, pairing geometry/time, controls and gate hierarchy.	Compare matched controls and verify the gate denominator before planning scale. [S1 p.1]
High apparent pairing; poor specificity	Single-population controls, compensation, nonspecific background, aggregates and multiple constructs per cell.	Retain ambiguous events and assignments; investigate what contributes to the signal.
Loss before capture	Pre/post-sort composition, handling, confirmed stabilization conditions, sorter rate and elapsed time.	Locate the loss by stage. A flow-only pilot does not measure downstream recovery. [S1 p.1; S8 p.1]

TO COMPLETE - C05

Resolve the BS3 temperature conflict and complete the stock/quench, wash and hold specifications. Use the confirmed version when investigating stabilization or recovery; do not choose between conflicting conditions by inference.

06 Target recovery and sequencing

Observation	First checks	Record or next action
HLA PCR2 fails	CZP31 versus CZP31.0 and the actual construct-binding site.	Resolve the documented obsolete-primer conflict before altering cycling. [S1 p.2; S10 p.1]
TCR PCR2 fails	Exact construct, CZP32UNI compatibility, template allocation and full PCR program.	Do not substitute CZP160 or a TCRAFT cloning primer without its validated compatibility record. [S10 p.1]
Short TCR-associated product	Whether it is the documented TCRAFT-derived dead-end branch or a different product.	Confirm identity, retain the original trace and use only the construct-specific cleanup. [S1 pp.2-3]
Unexpected fragment size/concentration	Primer coordinates, off-target products, cleanup, dilution correction and assay range.	Recalculate from raw measurements; distinguish contaminating DNA from target material. [S1 pp.2-3]
Low or imbalanced sequencing yield	Pool calculation, branch indexes, library QC, selected kit/flow cell and acquisition settings.	Separate loading, demultiplexing and assignment losses. More run time may not correct library-preparation errors. [S1 p.3]

TO COMPLETE - C10
Supply expected fragments from actual construct/primer coordinates, QC criteria, cleanup assays and the pooling calculation with its confirmed mass, molar or volume basis. Preserve all four branch measurements before pooling.

07 Assignments and interaction evidence

Observation	First checks	Record or next action
Multiple identities in a partition	Reference uniqueness, contamination, multiple constructs/cells, chimeras and barcode-processing rules.	Apply the released ambiguity policy. Do not select the strongest identity without a supported rule.
Expected association absent	Whether both identities were represented and recoverable, control behavior and sufficient sampling.	Separate detection/assignment failure or insufficient evidence from an interaction decision.
Same barcode in different reactions	Sample/channel identity and demultiplexing record.	Keep barcode namespaces separate. Matching strings across independent captures do not establish a pair.
Large read count but uncertain support	PCR amplification, molecule-identifier retention and the number of supporting partitions.	Report read support as reads unless independent molecule support is defined; retain ambiguity.

TO COMPLETE - C12
Release the sample/read architecture, parsing and assignment rules, background model, normalization, score/decision definitions, thresholds, replicate policy and executable example. These methods are not supplied by the bench notes.

Preserve unobserved, ambiguous, failed-QC and insufficient-evidence states where appropriate. An undetected construct is not a verified non-interaction. Keep experimental evidence and model predictions in separate fields or artifacts.

Close the record

Link the final result to protocol and material versions, instrument settings, FCS/gate files, fragment traces, pool calculations, sequencing files, checksums, code/reference versions and analysis configuration. Retain exclusions and failed runs with reasons. A clarification or changed condition belongs in the deviation record and method change log.

Source notes

Sources are the method authors' laboratory records, as transcribed in the supplied consolidation. The internal source files are not included in this download.

D1. CLASP-seq: the Cell_x_Cell TCR-pMHC pairing protocol. Consolidated author-review draft, 11 September 2026, sections 2.4, 8-10 and 12. The control plan, diagnostic records and completion requirements are proposed additions for reproducibility; they are not reported validation results.

S1. Cell_x_Cell TCR_HLA-pep Pairing Assay V1. Author-supplied laboratory notes, 4 pages; workflow and source-specific recovery/QC observations, as recorded in the consolidated draft.

S4. Generating Cell Lines for NxN Pairing Assay. Author-supplied laboratory notes, 2 pages; cell generation and representation.

S5. Spinfection for Cell_x_Cell. Author-supplied laboratory notes, 3 pages; transduction, recovery and selection.

S7. Violet cell tracer staining. Author-supplied laboratory notes, 1 page; incomplete tracer-stock specification.

S8. BS3 Crosslinking. Author-supplied laboratory notes, 1 page; stabilization and pre-sort handling.

S10. CZ PRIMERS AND GENEBLOCKS - CZP1-. Author-supplied primer registry, 2 pages; primer compatibility and deprecation notes on page 1.

Protocol and resources